MIL-S-40104B
5 February 1968
SUPERSEDING
MIL-S-40104A
23 September 1963

MILITARY SPECIFICATION

STRAINER, POT, CORROSION-RESISTING STEEL

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of corrosion-resisting steel pot strainer.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-W-423	-	Wire, Steel, Corrosion-Resisting
RR-W-360	-	Wire Fabric, Industrial
PPP-B-636	-	Box, Fiberboard

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of U.S. Military Property

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 7330

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2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL SANITATION FOUNDATION (NSF)

Standard No. 2 - Food Service Equipment (1965)

Seal of Approval Listing of Food Service Equipment

(Application for copies should be addressed to the National Sanitation Foundation, P.O. Box 1468, Ann Arbor, Michigan 48106.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

Standard A 167-63 - Corrosion-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Preproduction sample.- Unless otherwise specified (see 6.2), before production is commenced, a sample pot strainer shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

3.2 Standard product.- The pot strainer delivered under this specification shall be the manufacturer's commercially current standard product, except for any changes necessary to comply with the specification requirements.

* 3.3 Standard compliance.- The pot strainer covered by this specification shall conform to the applicable requirements of NSF Standard No. 2.

* 3.3.1 Certification.- Prior to approval of the preproduction sample, if one is submitted, or prior to the approval of the first shipment, the supplier shall submit satisfactory evidence to the contracting officer or his authorized representative that the pot strainer he proposes to supply

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under this specification meets the applicable requirements of NSF Standard No. 2. Acceptable evidence of meeting the requirements of the standard shall be the NSF seal, a listing in the NSF "Seal of Approval Listing of Food Service Equipment", or a certified test report from a recognized independent testing laboratory acceptable to the medical service of the procuring agency indicating that the pot strainer offered conforms to NSF Standard No. 2.

3.4 Materials.- The materials shall be new and of the type and quality normally furnished in the manufacturer's standard pot strainer, except as specified herein.

3.4.1 Corrosion-resisting steel.- The corrosion-resisting steel sheets shall have a chemical composition conforming to any of the 300 series of ASTM A 167-63.

3.4.2 Corrosion-resisting steel wire.- Corrosion-resisting steel wire shall conform to any composition in the 300 series, condition A of QQ-W-423.

3.4.3 Cloth wire.- Wire cloth shall be type 1, class 1, 6 by 6 mesh, 0.080-inch diameter, type 302 stainless steel, conforming to RR-W-360.

* 3.5 Design and construction.- The pot strainer shall be made of the corrosion-resisting steel specified herein and shall consist essentially of a handle, handle support, hook support, sieve, and a base. The body, base, and sieve shall be constructed as specified in 3.5.1 and 3.5.2 or shall be the alternate construction specified in 3.5.3. Dimensions shall be as specified in table I.

TABLE I.- Dimensions of pot strainer

Component	Dimensions
Body	Inches
Thickness	0.025 (24 gage) or thicker
Top diameter	9-3/4 ± 1/8
Bottom diameter	6-1/2 ± 1/8
Height	4-1/8 ± 1/8
Base	
Thickness	0.025 (24 gage) or thicker
Top and bottom diameter	6-1/2 ± 1/8
Height	1-3/8 ± 1/8
Sleeve diameter	6-3/8 ± 1/8

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TABLE I.- Dimensions of pot strainer (cont'd)

Component	Dimensions
Handle	
Thickness	0.085 nominal
Width	1-1/8 ± 1/8
Length	8-3/16 ± 1/8

3.5.1 Body and base.- The pot strainer body and base each, shall be of not more than two-piece construction with either double seamed joints, fillet welded or soldered on the interior, or shall be continuous seam welded. The top of the body shall have a rolled bead. The bottom of the body shall form a double seam with the sieve. The base shall then be fitted over this joint. In the area where the sleeve of the base meets the body, solder or fillet weld shall be applied. The bottom of the base shall have a wire reinforced bead.

3.5.2 Sieve.- The sieve shall be made of material specified in 3.4.3 or from material specified in 3.4.1 nominal 0.0187-inch thick, perforated by 0.085-inch diameter, plus or minus 0.005 inch, holes with a minimum of 36 holes per square inch. The sieve shall be flat and supported by two nominal 1/8-inch diameter cross wires, fabricated from material specified in 3.4.2. The cross wires shall be spot welded or soldered to the strainer body.

* 3.5.3 Alternate design for body, base, and sieve.- The pot strainer body and sieve shall be of one piece and seamless drawn from one piece of metal. The bottom of the body shall be perforated to form a sieve by 0.085-inch-diameter, plus or minus .005 inch, holes with a minimum of 250 holes per one quarter section of bottom. The top of the body shall have a one-half round open bead. The base shall be made from one piece of metal either seamless drawn or welded and ground smooth. The base shall then be welded to the body. The bottom of the base may be cut out to form four equally spaced legs.

3.5.4 Handle.- The handle shall be made of material specified in 3.4.1 and shall be shaped concave in cross section for added strength. The handle shall be provided with a hole for hanging. The handle shall be attached by spot weld, seam-weld, fillet weld, or solder.

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3.5.5 Hook support.- The hook support shall be made of material specified in 3.4.1 and shall be soldered, spot welded, seam-welded, or fillet welded to the upper edge of the strainer body opposite the handle. The periphery of the hook support joint shall have a solid bead of solder. The hook support shall be not less than 1 inch wide and 1 inch long.

3.6 Finish.- All sharp edges, burrs, and slivers shall be removed from the finished item. All rust, pits, and stains shall be removed.

3.7 Marking for identification.- The pot strainer shall be marked in accordance with MIL-STD-130, with the letters "US" and the manufacturer's name, trade name, or trademark.

3.8 Workmanship.-

3.8.1 Welding.- Welding shall be accomplished in a manner which shall prevent the occurrence of burn holes, cracks, fracture, or incomplete fusion. All scale or flux deposits shall be removed from finished welds.

3.8.2 Soldering.- Joints to be soldered shall be completely filled and wiped smooth after soldering. No flux shall remain on the joint or surrounding area after soldering is completed.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction sample inspection.- When a preproduction sample is required, it shall be examined for defects listed in table II and dimensions specified.

* 4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

* 4.3.1 Component and material inspection.- In accordance with 4.1, components and material shall be inspected and tested in accordance with all requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

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4.3.2 End item inspection.- The lot shall consist of all strainers offered for inspection at one time. The sample unit shall be one completely fabricated strainer.

* 4.3.2.1 Visual examination.- Examination shall be made of the strainer for defects listed in table II. The inspection level shall be II with an acceptable quality level (AQL) of 4.0 for major defects and 10.0 for total defects, expressed in terms of defects per hundred units.

TABLE II.- Classification of visual defects

Examine	Defect	Classification Major	Minor
Finish	Any burrs, sliver, splinter, or sharp edge	X	
	Rust, pits, and stains not removed		X
Construction and workmanship	Part missing or not specified type	X	
	Split, cracked, dented, or malformed		X
	Beading omitted or not as specified	X	
Body and base (as applicable to 3.5.1)	Each constructed with more than two pieces		X
	Not joined as specified	X	
Sieve (as applicable to 3.5.2)	Not flat		X
	Not supported as specified	X	
	Not secured as specified	X	
Body, base, and sieve (as applicable to 3.5.3)	Body and sieve not one piece seamless	X	
	Base not one piece	X	
	Holes in sieve not as specified and not deburred	X	
	Bead not as specified	X	
	Welds not ground smooth	X	
Handle	Not concave	X	
	Hole missing	X	
Hook support	Not constructed as specified	X	
	Not attached as specified	X	

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TABLE II.- Classification of visual defects (cont'd)

Examine	Defects	Classification Major	Minor
Construction and workmanship (cont'd)			
Welding	Missing, incomplete, burn holes, cracked, fractured, or otherwise not fused	X	
	Not a clean finish	X	
	Slag inclusion, slight undercut, not smooth and uniform, scale or flux deposits not removed		X
Soldering	Not adherent, incomplete, not clean (flux residue) not smooth, or pin holes in solder		X
Identification marking	Missing, incomplete, incorrect, not legible		X

4.3.2.2 Dimensional examination.- Examination shall be made of the completed strainer to determine compliance with dimensional requirements. Any dimension not within the specified requirement shall constitute a defect. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

* 4.3.3 Code and standards compliance.- Proof of compliance with the requirements of NSF (see 3.3 and 3.3.1) shall be available to the Government representative.

4.3.4 Examination of preparation for delivery.- An examination shall be made to determine that packing and marking as required by section 5 of this specification are complied with. Defects shall be scored as specified in table III. The sample unit shall be one shipping container fully prepared for delivery. The lot shall consist of the number of containers offered for delivery at one time. The inspection level shall be S-2 and the AQL shall be 6.5 defects, expressed in terms of per hundred units.

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TABLE III.- Examination of preparation for delivery

Examine	Defect
Marking, exterior and interior	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Component missing or damaged
Contents	Number per container is more or less than required

4.3.4.1 Examination of shipping containers .- When shipping containers are required to be in accordance with PPP-B-636, examination for defects in closure, waterproofing, and strapping shall be in accordance with PPP-B-636.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.- Each strainer shall be completely wrapped with minimum 30-pound basis weight kraft paper (24 x 36 - 500) and nested in units of three each. Six strainers shall then be packaged in a snug-fitting fiberboard box conforming to style RSC, type CF, variety SW, or type SF, class domestic of PPP-B-636. The strainers shall be positioned within the box with the bodies of one nest of strainers adjacent to the handles of the other nest, handles in diagonally opposite corners of the box, and one nest in the inverted position. Each box shall be closed in accordance with the appendix of PPP-B-636.

5.1.2 Level C.- The strainers shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing.- Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.- Twenty-four strainers, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, V2s of PPP-B-636. Each shipping container shall be closed, waterproofed by means of tape, and reinforced with flat strapping or tape banding in accordance with the appendix of PPP-B-636.

5.2.2 Level B.- Unless otherwise specified (see 5.2.2.1), twenty-four strainers, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF, variety SW, or type SF, class domestic of PPP-B-636. Each shipping container shall be closed in accordance with method II of the appendix of PPP-B-636.

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5.2.2.1 When specified (see 6.2), the shipping container shall be V3c or V3s, fabricated and closed in accordance with PPP-B-636.

5.2.3 Level C.- Strainers, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 Marking.- In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The pot strainer is intended for straining food and liquids in fixed messes, field kitchens, and galleys.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When a preproduction sample is not required (see 3.1).
(c) When V3c or V3s fiberboard is required for level B shipment (see 5.2.2.1).
(d) Selection of applicable level of packaging and packing (see 5.1 and 5.2).

6.3 The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - SA
Air Force - 84

Preparing activity:

Army - GL

Project No. 7330-0236

Review activities:

Army - MD
Navy - SA
Air Force - 84, 85

User activity:

Navy - MC

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS. This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

Strainer, Pot, Corrosion-Resisting Steel - MIL-S-4010 4B

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 56 WHICH MAY BE USED.

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